image recognition, handwritten digit classification, recurrent neural networks, NLP, word embedding and PDE (partial differential equations).

It supports a variety of different toolkits for constructing models at varying levels of abstraction. It has a flexible architecture with which it can run on a variety of computational platforms — CPUs, GPUs and TPUs.

Keras: This is used for constructing neural networks and for ML projects. It interoperates with TensorFlow, Theano and R. It runs efficiently on CPUs and GPUs.

Keras offers standalone modules including optimisers, neural layers, activation functions, initialisation schemes, cost functions, and regularisation schemes. It functions as a user-friendly, extensible interface that enhances modularity and total expressiveness.

PyTorch: This supports computer vision, ML and natural language processing. It integrates well with the Python data science stack, including NumPy.

It has a robust framework to build computational graphs on the go and even change them at runtime. It supports Tensor computation with the GPU, and helps with performance optimisation and scalable distributed training in research as well as production.

Pandas: This offers a wide range of tools for data manipulation and analysis. Using this library, we can read data from various data sources like CSV, SQL databases, JSON files and Excel. It can handle tabular data, time series data, matrix data, etc, and structure it into two types of data structures — data frames and series. Pandas is highly stable, providing an optimised performance.

Industry use cases of Python and ML

Healthcare: ML helps to monitor and predict diseases through applications, detects injuries, scans health parameters of patients regularly, etc. It can improve research, discover new drugs and help diagnose diseases more accurately. The ML technology has plenty of non-clinical uses also, such as the automation of administrative tasks and patient management systems.

Machine learning is also used for clinical trials of molecules, during epidemics like COVID-19, the discovery of new drugs, patient risk identification, etc.

Banking and finance: Fintech industries stand to gain much by using AI and ML applications because the latter can help in customising the user experience and detecting fraud — the two major concerns of this industry.

In addition, ML can be used in detecting customer attrition, loan defaulter prediction, fraud detection, credit scoring, money laundering prevention and portfolio management.

E-commerce: AI and ML based solutions help businesses to understand the shopper's needs and create a better customer experience, thus increasing sales revenue. ML helps in content personalisation, chatbots for improving performance, dynamic pricing, identifying shoppers' data patterns and predicting how responsive they might be to new prices.

Insurance: Python can help insurance companies in providing robust solutions for risk management, fraud management, personalised services, automation, customer support, etc.

Business services: Python is widely used for data gathering, predictions and ML tasks by business services.

Machine learning is used for chatbot creation in all the above sectors, personalising the customer experience and optimising labour usage.

The innovative services created with machine learning are already disrupting the markets. Python developers have kept pace by coming out with the latest libraries, frameworks and approaches to solving industry problems. These approaches enable businesses to use advanced ML techniques easily, to better their performance and improve the customer experience. END

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